Morning SPY hedge assessment

Report date: 2026-10-30

Verdict

The 2025 profit came mainly from favorable moves in the stock left overnight. The morning-price rule remains unproven as a repeatable strategy after costs.

The 2023 replication loses money, including over the same January-August window. All estimates retain the frozen theoretical daily-close design.

Why the years differed

The overnight stock-price term explains 98.8% of the $702.35 mean P&L gap. Residual short-stock positions contribute most, benefiting from larger overnight declines in 2025.

The same January-August window still loses in 2023. Removing 2025's ten best trades turns its mean negative; removing five leaves a smaller profit.

With the overnight price term removed and retained modeled costs held fixed, both years lose. This accounting check does not model a risk-free hedge.

USD per package of ten short SPY option contracts and one long SPXW option contract. Figures are dollar P&L, not rates or annualized returns. Primary comparison uses full-cross, base costs, gross prefunding, and 1 bp stock slippage.

Main comparison

Period	n	Mean	Median	Total	95% CI	Win rate
2023 calendar year	196	-$48.96	-$66.70	-$9,596.60	[-$221.01, $117.67]	40.3%
2025 Jan-Aug	127	$653.39	-$10.08	$82,980.62	[$125.64, $1,274.85]	48.0%

2023 source coverage

Coverage check	Count
Expected pairs	196
Complete pairs	196
Pair/fill rows	588
Scenario cases	35280
Actual returned quote rows	68903
Projection fields matched	29

The statistical denominator is 196 pairs. The 588 pair/fill rows represent three hypothetical fill variants per pair, not 588 independent quote samples. The 35,280 scenario cases apply scenario combinations to those rows.

The 2023 replication is an additional historical sample, not a globally untouched or predictive holdout. The 2025 comparison covers January through August only, not a full year, and is not a paired inference against 2023.

Attribution: cohort comparison

2023 full year and January-August use different windows. The January-August comparison holds the calendar window constant. These are historical comparisons, not a predictive holdout.

Cohort	n	Mean USD	Median USD	Total USD	Win rate	Min expiry	Max expiry
2023 full year	196	-$48.96	-$66.70	-$9,596.60	40.3%	2023-01-04	2023-12-29
2023 Jan-Aug	131	-$56.12	-$47.16	-$7,351.92	41.2%	2023-01-04	2023-08-31
2025 Jan-Aug	127	$653.39	-$10.08	$82,980.62	48.0%	2025-01-03	2025-08-28

Mean contribution diagnostics, USD per package.

Cohort	Overnight mean	Option baseline	Dividend	Mean costs	Net excluding overnight	Winsorized 5-95 mean
2023 full year	$82.96	-$8.48	-$16.04	$107.39	-$131.92	-$40.24
2023 Jan-Aug	$71.07	$1.42	-$24.00	$104.61	-$127.19	-$48.99
2025 Jan-Aug	$777.20	$18.45	$13.87	$156.13	-$123.81	$604.60

Additive mean-P&L bridge, USD per package. Confidence intervals describe component differences. Costs appear as negative contributions.

Component	2023 mean	2025 mean	Difference	95% CI low	95% CI high
Option baseline	-$8.48	$18.45	$26.93	-$103.72	$116.26
Overnight stock contribution	$82.96	$777.20	$694.25	$150.03	$1,329.92
Dividend cash	-$16.04	$13.87	$29.91	$0.00	$70.81
Option fees (negative)	-$7.15	-$7.15	$0.00	$0.00	$0.00
Stock fees (negative)	-$4.34	-$4.84	-$0.51	-$1.24	$0.22
Short borrow (negative)	-$10.72	-$10.48	$0.24	-$5.83	$5.86
Premium debit funding (negative)	-$0.01	-$0.01	$0.00	-$0.01	$0.00
Hedge funding (negative)	-$48.03	-$75.75	-$27.71	-$44.48	-$11.15
Stock slippage (negative)	-$37.14	-$57.89	-$20.75	-$29.16	-$12.19

Net excluding overnight keeps retained modeled costs fixed as an accounting diagnostic. It does not model a risk-free hedge strategy. A favorable overnight stock contribution is cash flow attribution, not evidence that the rule predicts or causes profit.

Attribution: residual and tail checks

Residual frequency and payoff effects use a symmetric decomposition. Tail rows remove fixed counts from each cohort separately; they are post-hoc checks, not tradable filters. Winsorized means are standalone summaries and do not add across components.

All dollar amounts are USD per package. Year labels refer to 2023 and 2025. p is the cohort share. Conditional values are mean overnight contributions per residual state; all-cohort values are weighted overnight contributions.

State	n '23	n '25	p '23	p '25	Cond. '23	Cond. '25	All '23	All '25	Frequency	Payoff
short	36	21	18.37%	16.54%	-$15.83	$3,155.24	-$2.91	$521.73	-$28.76	$553.40
flat	127	79	64.80%	62.20%	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
long	33	27	16.84%	21.26%	$510.00	$1,201.67	$85.87	$255.47	$37.85	$131.75

Tail removal uses fixed counts. Original denominators remain 196 for 2023 and 127 for 2025.

Case	n 2023 after	mean 2023 after	removed sum 2023	n 2025 after	mean 2025 after	removed sum 2025
Remove best 1	195	-$74.66	$4,962.95	126	$497.67	$20,273.94
Remove worst 1	195	-$27.90	-$4,156.44	126	$736.85	-$9,862.21
Remove best 5	191	-$143.47	$17,805.39	122	$152.59	$64,364.26
Remove worst 5	191	$43.58	-$17,920.47	122	$981.57	-$36,771.22
Remove best 10	186	-$218.13	$30,975.94	117	-$105.30	$95,300.95
Remove worst 10	186	$122.34	-$32,351.80	117	$1,117.78	-$47,799.29

Common January-August monthly comparison, USD per package.

Month	n 2023	mean 2023	total 2023	n 2025	mean 2025	total 2025
01	15	-$317.49	-$4,762.35	14	-$107.17	-$1,500.41
02	15	$207.01	$3,105.16	15	$816.54	$12,248.17
03	19	-$291.41	-$5,536.82	16	$859.12	$13,745.88
04	15	$660.30	$9,904.54	17	$1,017.19	$17,292.28
05	17	-$165.66	-$2,816.28	17	$1,612.49	$27,412.35
06	17	-$424.32	-$7,213.46	14	$784.83	$10,987.58
07	14	-$131.94	-$1,847.10	18	$171.21	$3,081.85
08	19	$95.49	$1,814.38	16	-$17.94	-$287.09

Return scales

Year	Mean expiry SPY close, USD	Mean absolute overnight return, %	Mean signed overnight return, %
2023	$427.22	0.389%	0.031%
2025	$593.56	0.573%	-0.037%

Year-gap intervals are unpaired descriptive intervals, not paired year inference.

Window	95% CI low	95% CI high
2023 full year vs 2025 Jan-Aug	$145.76	$1,356.61
Both Jan-Aug	$140.25	$1,378.36

Driver findings

The overnight stock-price term explains 98.8% of the $702.35 mean P&L gap. Residual short-stock positions contribute most, benefiting from larger overnight declines in 2025.

The same January-August window still loses in 2023. Removing 2025's ten best trades turns its mean negative; removing five leaves a smaller profit.

With the overnight price term removed and retained modeled costs held fixed, both years lose. This accounting check does not model a risk-free hedge.

2023 primary arms

All five arms use the same 196-pair sample. Absolute intervals measure each arm; paired intervals compare each arm with q0 on the shared sample.

Arm	n	Mean	Median	Absolute 95% CI	Paired vs q0 95% CI	Win rate
q0	196	-$124.54	-$17.15	[-$320.93, $75.75]	[$0.00, $0.00]	47.4%
fixed 500	196	-$89.95	-$150.68	[-$222.96, $45.40]	[-$124.66, $178.70]	44.4%
fixed 1,000	196	-$55.35	-$137.01	[-$267.47, $144.87]	[-$249.33, $357.40]	33.2%
morning-price rule	196	-$48.96	-$66.70	[-$221.01, $117.67]	[-$119.83, $273.89]	40.3%
assignment oracle	196	-$91.44	-$77.16	[-$114.53, -$70.07]	[-$164.47, $227.95]	32.7%

Legacy period means

The historical comparison preserves the reviewed period means. The 2023 sample is shown separately and is not paired with 2025.

Period	n	Mean	Median	Total	95% CI	Win rate
2019 pilot	9	-$610.75	n/a	n/a	n/a	n/a
2022 H2	63	-$136.85	n/a	n/a	n/a	n/a
2024	195	-$90.03	n/a	n/a	n/a	n/a
2025 Jan-Aug	127	$653.39	-$10.08	$82,980.62	[$125.64, $1,274.85]	48.0%
2023 calendar year	196	-$48.96	-$66.70	-$9,596.60	[-$221.01, $117.67]	40.3%

Risk, contribution, and costs

Residual position counts, overnight price contribution, costs, notional, and drawdown describe the primary morning-price scenario.

Long positions	Short positions	Nonzero positions
33	36	69

Residual share exposure

Measure	Mean shares	Total shares
Signed residual shares	-15.31	-3,000.00

P&L contribution, mean and total USD

Measure	Mean USD	Total USD
Overnight price contribution	$82.96	$16,260.00
Gross P&L	$58.43	$11,452.43
Net P&L	-$48.96	-$9,596.60

Cost components, mean and total

Component	Mean	Total
Hedge funding	$48.03	$9,414.52
Option contract fees	$7.15	$1,401.40
Premium debit funding	$0.01	$1.96
Residual short borrow	$10.72	$2,102.06
Stock transaction fees	$4.34	$850.00
Stock slippage	$37.14	$7,279.09

Notional and drawdown

Maximum notional	Maximum peak-to-trough drawdown
$476,880.00	$23,950.32

Month cohorts

Month	n	Mean	Median	Total	Long positions	Short positions	Nonzero positions
2023-01	15	-$317.49	-$104.88	-$4,762.35	1	4	5
2023-02	15	$207.01	-$7.15	$3,105.16	2	4	6
2023-03	19	-$291.41	-$120.33	-$5,536.82	2	3	5
2023-04	15	$660.30	$76.66	$9,904.54	4	3	7
2023-05	17	-$165.66	-$53.58	-$2,816.28	3	1	4
2023-06	17	-$424.32	-$86.77	-$7,213.46	2	6	8
2023-07	14	-$131.94	-$111.86	-$1,847.10	2	3	5
2023-08	19	$95.49	-$17.15	$1,814.38	7	2	9
2023-09	16	-$856.56	-$145.14	-$13,705.02	3	2	5
2023-10	17	$542.12	$62.85	$9,215.96	4	2	6
2023-11	16	$11.85	-$104.38	$189.59	2	2	4
2023-12	16	$128.42	-$147.43	$2,054.79	1	4	5

Morning rule sensitivity

The table varies fill convention, base or stress costs, stock slippage, and funding treatment for the fixed morning-price rule.

Funding labels: Gross pre-funding means gross_prefunding. Assignment cash net means assignment_cash_net.

Fill	Cost	bp	Funding	Mean net USD	Win rate
Midpoint	Base	0	Gross pre-funding	$5.83	48.0%
Midpoint	Base	0	Assignment cash net	$38.05	56.1%
Midpoint	Base	1	Gross pre-funding	-$31.31	43.9%
Midpoint	Base	1	Assignment cash net	$0.91	52.0%
Midpoint	Base	5	Gross pre-funding	-$179.86	34.7%
Midpoint	Base	5	Assignment cash net	-$147.64	37.8%
Midpoint	Stress	0	Gross pre-funding	-$80.92	38.8%
Midpoint	Stress	0	Assignment cash net	-$16.49	52.6%
Midpoint	Stress	1	Gross pre-funding	-$118.06	35.7%
Midpoint	Stress	1	Assignment cash net	-$53.63	46.9%
Midpoint	Stress	5	Gross pre-funding	-$266.61	31.1%
Midpoint	Stress	5	Assignment cash net	-$202.18	33.2%
Halfway to cross	Base	0	Gross pre-funding	-$3.00	45.9%
Halfway to cross	Base	0	Assignment cash net	$29.22	54.6%
Halfway to cross	Base	1	Gross pre-funding	-$40.13	42.3%
Halfway to cross	Base	1	Assignment cash net	-$7.92	50.0%
Halfway to cross	Base	5	Gross pre-funding	-$188.69	33.7%
Halfway to cross	Base	5	Assignment cash net	-$156.47	36.7%
Halfway to cross	Stress	0	Gross pre-funding	-$89.75	38.3%
Halfway to cross	Stress	0	Assignment cash net	-$25.32	52.0%
Halfway to cross	Stress	1	Gross pre-funding	-$126.89	35.2%
Halfway to cross	Stress	1	Assignment cash net	-$62.46	44.9%
Halfway to cross	Stress	5	Gross pre-funding	-$275.44	30.6%
Halfway to cross	Stress	5	Assignment cash net	-$211.01	32.7%
Full cross	Base	0	Gross pre-funding	-$11.82	43.9%
Full cross	Base	0	Assignment cash net	$20.39	52.6%
Full cross	Base	1	Gross pre-funding	-$48.96	40.3%
Full cross	Base	1	Assignment cash net	-$16.75	48.0%
Full cross	Base	5	Gross pre-funding	-$197.52	32.1%
Full cross	Base	5	Assignment cash net	-$165.30	34.2%
Full cross	Stress	0	Gross pre-funding	-$98.58	37.2%
Full cross	Stress	0	Assignment cash net	-$34.15	49.5%
Full cross	Stress	1	Gross pre-funding	-$135.72	34.2%
Full cross	Stress	1	Assignment cash net	-$71.28	43.4%
Full cross	Stress	5	Gross pre-funding	-$284.27	29.6%
Full cross	Stress	5	Assignment cash net	-$219.84	31.6%

Method and limitations

The fixed rule sets q to 1,000 shares when the official SPY open on the expiry session is at least the frozen entry-selected SPY strike plus $0.01; otherwise q is zero. The comparison tolerance is 1e-10. The morning price input chooses hedge size. It is not the modeled purchase price. The modeled hedge is bought at expiry close, then residual stock is closed at the next open. The 2023 quote clock is unverified. Some 2023 option examples and 2019–2025 stock history were inspected earlier; the entire new 68,903-row quote export was not. The 2023 sample is an additional historical replication, not a globally untouched or predictive holdout. The 2025 sample covers January through August only, not a full year, and is not paired inference against 2023. Daily theoretical fills do not prove execution, capacity, funding availability, or margin. Physical assignment is a proxy; early assignment remains possible. Do not extend the 2024/2025 15:59 quote claim to 2023. Cboe actual SPXW PM settlement is not ordinary SPX close. The oracle uses future assignment information and is diagnostic only, not a tradable rule or profit bound. No tuned rule, profit guarantee, or trading recommendation is implied.

Costs: 11 option contracts at $0.65 base / $1.50 stress each; stock fees $0.005 / $0.01 per share; annual borrow 3% / 10%; funding 5% / 10%; SPY and SPXW option contract multipliers are 100; package ratio is ten SPY options short to one SPXW option long. Bootstrap uses shared expiry ISO week clusters, 10,000 draws, seed 20261030, linear type-7 quantiles.

Sources: WRDS queries 11847843 and 11847851; Cboe W/M/S 2023 settlement files https://www.cboe.com/index_settlement_values/get_sv_data/W/2023/, https://www.cboe.com/index_settlement_values/get_sv_data/M/2023/, https://www.cboe.com/index_settlement_values/get_sv_data/S/2023/.

Provenance

Validation PASS, 1777327 checks, receipt SHA-256 `4e297cb6a8c941c9dce33532c7355aad4d3ce0d39d30a105d42aeab1e4b0cba3`; max error 5.2523319027386606e-11.

Git 64e72887b782fdf838257a09877ffb817dfbfbdf clean=True; displayed command (private paths redacted): [local path] -m spyspx.morning_hedge_2023 evaluate --input-manifest [local path] --input-manifest-sha256 `47972793ea9daadfa35e5fa7aea884f61759d750525384946f18994dc1b89656` --selection [local path] --selection-sha256 `00cff4fb8c3ec13d9ea72cc4ee4d4c5cc1723747159bdea723305e2c6a167f13` --out planning/morning-hedge-replication-20261030/evaluation-01. The exact executed command is retained in the bound private execution receipt.

Input manifest `47972793ea9daadfa35e5fa7aea884f61759d750525384946f18994dc1b89656`; protocol `8f66787ce63d47771c1bbb5edd516c02d46ebc37b2538f3581ae3e073cbfa403`; calendar `122296e11825bbdec9af5bbed5cca5f3fc6c8ea0dc99548351dc05d44beee809`; selection `00cff4fb8c3ec13d9ea72cc4ee4d4c5cc1723747159bdea723305e2c6a167f13`; adapter output `6cf57c05e5ab8151d2e711fbff39437a93b821320def08141565460a15f3bb8c`; execution `3b50a9eea10ee0574f9fb2f6162ff491463503c50107415448d34a9865cc766a`.

Adapter code hashes: code_adapter=`02cd50de29e5c843a2bbace14ce1825d6027ea2268eec105878396a2030b8a13`, code_admission=`c9ab3d063535b588ab0f9d42f2d8549e6f6d711942a6aff2cc2cca9f8439a506`, code_expiry_extension=`6a5964a78334cbe2d8fa3f1285221223eb609daeb39061036e62a022f9c481d0`, code_expiry_payoff=`c7eaf0c90fd5e029dac3012630af88646c0c8329697279a198ffd4842a8fbb48`, code_normalizer=`045971a5b41c83d311b40f6733a7c20578b5e326ada4fdbc0417a9e1506a8130`, code_overnight_hedge=`6cca6772325e7d55c8a85ef6a444681d2a54175e9ddfb10949dfdae6748337f3`, code_replication=`c010d765e58d0130b3f61b4253a8e1e08ca068b5e7a920c69e424e12952e59b8`.

Legacy statistics `295078e0fa11b339cfa995786b9536ca8240c8ff98c9911f33684ec69ccfcb55`; render input `1d5702022d60cb0bf89e50aaece1b2c2d2c762704c580400c6c02a5c7305881e`; renderer `da424fee394728e081449afbd9e3cca7cf7234ab2bba943cb77524d5439a9d23`; PDF recorded in Markdown and render receipt.

Attribution run appendix

Attribution intervals are uncorrected, post-hoc 95% descriptive intervals from 10,000 independent-year ISO expiry-week cluster resamples. The full-window comparison uses seed 20261030 and the January-August comparison seed 20261031. Each draw uses sampled row-sum totals divided by sampled row counts, with type-7 quantiles. They are unpaired and do not establish predictive or causal effects.

Attribution run SHA 9a7c219c6ef886929621fc3863cf51dca660a6d6, clean=True, started 2026-10-30T05:29:00.099609+00:00, ended 2026-10-30T05:29:02.697712+00:00. The report's original 2023 evaluation SHA remains 64e72887b782fdf838257a09877ffb817dfbfbdf; the 2025 source evaluation SHA is db5d980c7abc344af02e8ae98fc426da92df433e.

Independent validation PASS, 2189579 checks, 0 mismatches, max error 8.585629984736443e-10; checker independent raw-source cash-flow and bootstrap checker (`b3b4029c6536ab1903d8433a9f22d721 5c3f1e0f04252d3cef149d36a1930c80`); receipt `011edba0f0bd7949e78ac1662551945d 1298d1b9f8a8b3ece5c048f7372558bf`.

Command text and per-file source and output hashes are recorded in the matching Markdown report; local paths are redacted.